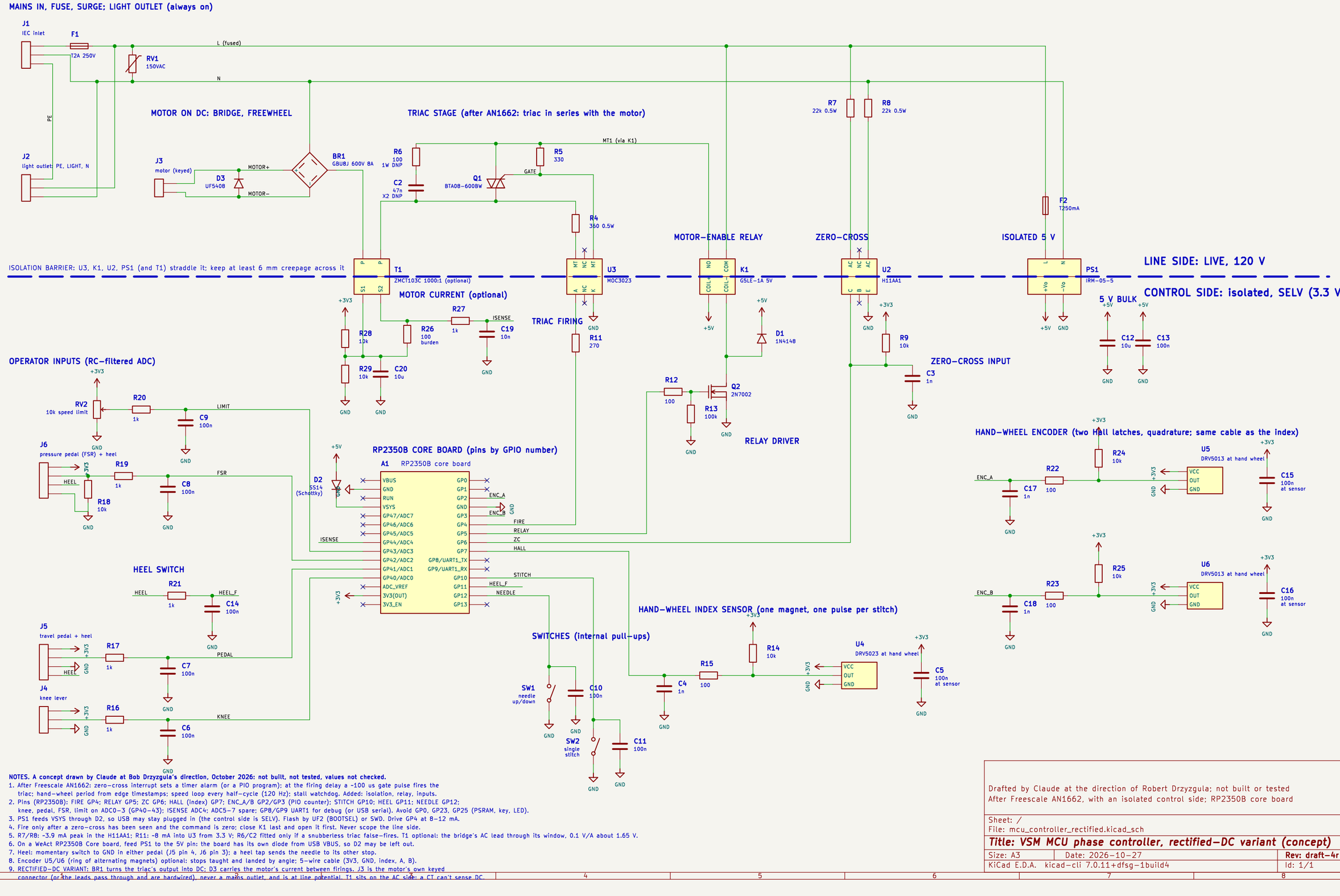


NOTES. A concept drawn by Claude at Bob Drzyzgula's direction, October 2026: not built, not tested, values not checked.

1. After Freescale AN1662: zero-cross interrupt sets a timer alarm (or a PIO program); at the firing delay a ~100 us gate pulse fires the triac; hand-wheel period from edge timestamps; speed loop every half-cycle (120 Hz); stall watchdog. Added: isolation, relay, inputs.
2. Pins (RP2350B): FIRE GP4; RELAY GP5; ZC GP6; HALL (index) GP7; ENC_A/B GP2/GP3 (PIO counter); STITCH GP10; HEEL GP11; NEEDLE GP12; knee, pedal, FSR, limit on ADC0–3 (GP40–43); ISENSE ADC4; ADC5–7 spare; GP8/GP9 UART1 for debug (or USB serial). Avoid GP0, GP23, GP25 (PSRAM, key, LED).
3. PS1 feeds VSYS through D2, so USB may stay plugged in (the control side is SELV). Flash by UF2 (BOOTSEL) or SWD. Drive GP4 at 8–12 mA.
4. Fire only after a zero-cross has been seen and the command is zero; close K1 last and open it first. Never scope the line side.
5. R7/R8: ~3.9 mA peak in the H11AA1; R11: ~8 mA into U3 from 3.3 V; R6/C2 fitted only if a snubberless triac false-fires. T1 optional: the bridge's AC lead through its window, 0.1 V/A about 1.65 V.
6. On a WeAct RP2350B Core board, feed PS1 to the 5V pin: the board has its own diode from USB VBUS, so D2 may be left out.
7. Heel: momentary switch to GND in either pedal (J5 pin 4, J6 pin 3); a heel tap sends the needle to its other stop.
8. Encoder U5/U6 (ring of alternating magnets) optional: stops taught and landed by angle: 5-wire cable (3V3, GND, index, A, B).
9. RECTIFIED-DC VARIANT: BR1 turns the triac's output into DC; D3 carries the motor's current between firings. J3 is the motor's own keyed connector (or the leads pass through and are hardwired), never a mains outlet, and is at line potential. T1 sits on the AC side; a CT can't sense DC.

Drafted by Claude at the direction of Robert Drzyzgula; not built or tested
After Freescale AN1662, with an isolated control side; RP2350B core board

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File: mcu_controller_rectified.kicad_sch		
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